CS 641: Modern Cryptography
04 May 2026, 5:00 – 8:00 pm

End Semester Exam
Total marks: **30**

Answer each of the following questions with justifications. No marks will be awarded without correct reasoning.

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1. Diffie-Hellman Key Exchange (DHKE) allows sharing a secret keys between 2 parties. Show how can you use DHKE to share the same secret key among 3 participants. ☒

2. Using Extended Euclidean Algorithm, compute 33^{-1} mod 2026. (Calculators are not permitted.) ☒

3. Prove that collisions can be found for a cryptographic hash function with n bit output in $\mathcal{O}(2^{n/2})$ calls to the hash function. ☒

4. Let $e : G_1 \times G_1 \to G_2$ be a symmetric bilinear pairing for some suitable groups G_1 and G_2. Prove that DDH problem is easy in G_1. ☒

5. Your friend is learning to design Block ciphers and suggests the following constructions. Explain why or why not these are good constructions. In both the constructions below, K is the secret key, R is a randomly generated n-bit string, and M is the message. Further, $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ is a secure PRF. However, F is not a PRP. ☒

   $15 \times 4^{-1} \bmod 7$

   (a) $\mathsf{Enc}(K, M) = (R, F(K, R) \oplus M)$

   1×2

   (b) $\mathsf{Enc}(K, M) = (R, F(K, M) \oplus R)$

6. For the elliptic curve $E : y^2 \equiv x^3 + 3x + 4 \pmod 7$, and a point $P = (2, 2)$ on E, compute $2P$. $\boxed{3}$

   $4 \quad 8+6 \quad 18$

7. Let $G : \{0,1\}^n \to \{0,1\}^{2n}$ be a PRG. Is the following construction a PRG? Justify your answer (either show that the construction satisfies the definition of a PRG or show a distinguisher against it). ☒

   $$G'(s) = a||b||c||d \text{ where } G(s) = a||b, \text{ and } G(b) = c||d$$

   Note that $a, b, c, d \in \{0,1\}^n$ and $||$ denotes concatenation.

8. Let $n = p \cdot q$ for large primes p, q, and (e, d) be the usual parameters for textbook-RSA. You decide to digitally sign a message m using the RSA signature algorithm (sign $s = m^d \bmod n$). But to speedup the computation, you decide to use computations mod p and mod q separately, and then combine the results using the Chinese remainder theorem. $\boxed{3}$

   Your computation goes correctly mod p but, somehow, your computer corrupts the computation mod q. This incorrect computation leads to an incorrect signature s' for the message m. Show that an attacker who observes the message m and the incorrect signature s' can factor n. (The attacker knows the public values n, e but does not know d. Further, the attack succeeds irrespective of what mistake is made mod q. The attacker only knows the fact that the computation was correct mod p but wrong mod q.)

9. Describe an attack on El-Gamal Cryptosystem using Shor's algorithm. How does it impact the security of this cryptosystem? ☒

10. We wish to share a secret $s = 7$ among 5 people such that whenever any 3 or more participants get together, they can recover the secret. But whenever 2 of them combine, they do not learn anything about the secret. We wish to work over Z_{11} for this. What should we share with the 5 participants? ☒

11. Suppose we have a cyclic group $G = \langle g \rangle$ and that $|G| = p$, a large prime number. Both g and p are public parameters. ☒

    Peggy (P) has a secret s and she wishes to prove the existence of this secret to Victor (V) in zero-knowledge. She first publishes her public key $y = g^s$. Then the following protocol is followed.

    $$\begin{aligned} P \to V &: \ a \text{ where } a = g^r \text{ for a random } r \\ V \to P &: \ \text{a random challenge } c \\ P \to V &: \ z \text{ where } z = (c \cdot s + r) \bmod p \\ &V \text{ accepts the proof if } g^z = a \cdot y^c \end{aligned}$$

    Prove that the above protocol satisfies (i) completeness, (ii) soundness, and (iii) zero-knowledge properties.

$4 \quad \cancel{177 \cdot 4 - 225} - 1054 \quad 708 \quad +16$

$225 \quad 420 \quad 645 \qquad 708 \quad 645 \quad 63 \qquad 15 \quad 15 \quad 165$

$\frac{3+165}{4} \qquad 12 \quad 165 \quad \frac{177}{4} \qquad 4 - \frac{121}{4} \qquad \frac{121}{16} \quad -105$